Firm Crosswalk Matching Report

Project: AI Workforce Statements
Date: October 23, 2026
Study period: 2018–2025

1. Data and Sample

- I built an initial historical S&P 500 firm crosswalk using CRSP, CCM, Compustat Fundamentals Annual, and Revelio company mapping data.
- The CRSP sample contains **664 securities (PERMNO)** and **657 company identifiers (PERMCO)**, covering **666 historical S&P 500 membership intervals**.
- I used historical CCM link dates rather than matching companies solely by their current tickers.

2. Matching Results

- **CRSP → CCM:** 657 of 664 securities (98.95%) have a date-overlapping primary P/C link. At the membership-interval level, 659 of 666 intervals (98.95%) have a P/C candidate, representing **652 unique GVKEY**.
- **CCM → Compustat:** 640 of 652 GVKEY (98.16%) have records in the downloaded Compustat annual dataset.
- **Compustat → Revelio:** 539 of 652 GVKEY (82.67%) have a **direct** GVKEY-based RCID candidate.
- Searching by CIK and full CUSIP finds candidates for **26 additional firms**: 25 supported by both identifiers and one by CUSIP only.
- Overall, **565 of 652 firms (86.66%)** have at least one Revelio candidate in the local extract. The remaining **87 firms** have no candidate under the current methods.

These are candidate coverage rates, not verified firm-level matches. The Revelio file was assembled from ticker-based queries and may not cover every company available in Revelio.

3. Unmatched Firms and Data Issues

- Seven S&P 500 membership intervals do not have a date-overlapping P/C link. Six have secondary J-link evidence, and one has a one-day date discrepancy.
- One additional interval has partial CCM coverage. Two P/C records use the special LS link type and remain flagged for review.
- Twelve candidate GVKEY are missing from the current Compustat annual extract.
- Among firms without any local Revelio candidate, the largest by available Compustat employment include **Berkshire Hathaway, Walgreens Boots Alliance, Marsh, Gap,**

Raytheon, WestRock, Nordstrom, Interpublic Group, Bank of New York Mellon, and Foot Locker.

- Employment rankings are provisional because reporting years differ across firms. Of the 87 firms without a local candidate, 13 have no dated numeric employment value available for ranking.

4. Crosswalk Output

I exported the first provisional version of the crosswalk:

- **`crosswalk.parquet`** — 652 GVKEY-level firm records, with available identifiers, candidate evidence, and review flags.
- **`crosswalk_membership_bridge.parquet`** — all 666 historical membership intervals, including the seven unresolved P/C cases.
- **`crosswalk_schema.md`** **and** **`crosswalk_log.md`** — field definitions, matching procedures, and outstanding decisions.

Both Parquet files passed the Notebook's read-back and consistency checks.

5. Next Steps

- Review the CIK/CUSIP identity checks and reconcile outstanding candidate-level review flags.
- Agree on a consistent rule for accepting GVKEY-to-RCID matches, especially where identifiers or parent-company relationships differ.
- Investigate the 87 firms without local Revelio candidates and determine whether additional company-mapping queries are needed.
- Use the approved crosswalk to connect earnings-call statements with Revelio workforce data.

The main crosswalk structure is in place. The remaining issue is deciding which Revelio identities can be accepted for the workforce analysis, rather than forcing matches for every firm.

Notebook:
https://github.com/kptransfer2024-cmd/ai-workforce-statements/blob/main/code/notebooks/01_build_firm_crosswalk.ipynb